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# Learning to Manipulate a New Object from Action-Free Videos

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Group 9  
McGill University  
ECSE 552 — Project proposal

## 1 Background

**Application and problem.** [TO WRITE: A1–A2, about 150 words. The application, the problem, and why a solution is needed.]

- Robot imitation learning needs demonstrations paired with motor commands, which are expensive to collect; videos without actions are much cheaper.
- Setting: action-labeled demonstrations with one object (cube) and only action-free videos with a new object (elongated block).
- Question: can these videos teach a policy to manipulate the new object?
- Labels are withheld in simulation on purpose, to emulate video-only data.
- Inverse-dynamics pseudo-labeling already exists (BCO, VPT), but assumes labeled and unlabeled data come from the same domain; here the object changes.
- Frame the contribution as a controlled study of adaptation across object geometry, not a new way of learning from video.

**Design constraints.** [TO WRITE: A3, about 80 words. Requirements the design must satisfy, not team time or equipment.]

- Target data are only ordered RGB images: no actions, robot states, object poses, or annotations.
- The learned policy sees only current and past images.
- It must succeed in closed loop, with a fixed budget of source labels.
- Simulator state is never an input to a learned model.

## 2 Aims/Goals

**Scope.** [TO WRITE: A4, about 80 words. What is in and out of scope.]

- OCBench visual single-block task in MuJoCo, UR5e arm with Robotiq gripper.
- Source: the existing cube. Target: a new elongated-block variant, which needs adapted expert grasps.
- Keep cameras, lighting, goal, and initial-position distribution fixed.
- Out of scope: human videos, real hardware, other spatial shifts.

**Hypothesis and contribution.** [TO WRITE: A5, about 70 words. The hypothesis and the expected contribution.]

- Domain-adversarial training of the IDM may give better target pseudo-labels, and therefore better policy success, than an unadapted IDM.
- Second question: does alignment help more in the IDM or in the policy?
- Risk to state honestly: invariance may also erase information needed to predict actions; a negative result is still informative.

### 3 Methodology

**Data and implementation.** [TO WRITE: A5–A6, about 200 words. Data generation, models, inputs and outputs, training objectives.]

- Generate demonstrations with OCBench’s scripted expert and data generator; capture an image at every control step (the published visual data are sparse).
- Actions: OCBench’s existing normalized joint and gripper increments.
- IDM: predicts the action from a short window of frames; it may look at future frames because it only labels data offline.
- Adaptation: a domain classifier (cube vs. block) with gradient reversal on the IDM encoder, while the action head keeps predicting source actions.
- Freeze the IDM, pseudo-label the block videos, train the policy on real cube labels plus block pseudo-labels.
- To decide: number of episodes, cameras, encoder, history length, loss weights.

**Alternatives.** [TO WRITE: A8, about 100 words. Alternatives run on our own data, and why ours could be better.]

- Source-only cloning (ignores the block videos).
- Pseudo-labels from an unadapted IDM (the BCO/VPT recipe).
- Alignment in the policy instead of the IDM.
- Reference trained with the true block actions (not a guaranteed upper bound).
- A published policy such as Diffusion Policy on the same data.
- Latent-action methods (LAPA, MotoVLA) as related work, too large to rerun.

**Validation and evaluation.** [TO WRITE: A7, about 100 words. Splits, model selection, metrics, episodes and seeds.]

- Split by episode before cutting windows: source validation, target adaptation, target development, final test.
- Never use hidden block actions to select the adapted model.
- Select policies by rollout success, since validation loss need not track task success (OCBench).
- Metrics: success rate per seed, failure type (grasp, transport, placement), action error, source retention.

### 4 Feasibility and resources

**Dataset, resources, and pilot.** [TO WRITE: A9, about 80 words. Dataset, compute, pilot, timeline.]

- Data are generated by us in OCBench on CPU MuJoCo; GPU access to confirm.
- Pilot checks: the expert succeeds on both objects, the source policy succeeds on the cube, success drops on the block, the reference learns the block.

### 5 Risk analysis

**Pitfalls and mitigation.** [TO WRITE: A10, about 80 words. What could fail and what we will do about it.]

- The block is too hard for the expert: shorten it, narrow its rotations.
- Source-only already succeeds on the block: make the geometry change larger.
- The cube demos do not cover the motions the block needs.
- Alignment hurts: report it as negative transfer.

## 6 Role of group members

[TO WRITE: Explicit division of work: who does what.]

- Possible work packages: simulation and data; policy and baselines; IDM and adaptation; evaluation and integration.

**Generative-AI disclosure.** Codex (OpenAI) and Claude Code (Anthropic) assisted with the document structure, starting ideas, literature search, bibliography, and LaTeX formatting. [TO WRITE: Confirm: the proposal text was written by the team.]

## References